Social-Worker Visits, Mutual-Aid Pairing and Volunteer Matching: A Layered Agent-Based Simulation of Social Support Interventions for Older Adults Living Alone

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Abstract

Loneliness and social isolation among older adults living alone are shared concerns of social work, social security and social governance. Communities support these elders through three main channels: professional social-worker visits, mutual-aid pairing of younger-old with older-old residents (organised through a time bank), and volunteer matching through a digital platform. Every service ends, and the hardest practical question is what is left afterwards: which layer of relationships has actually been changed? In the real world the same elders cannot both receive and not receive an intervention. We build a layered agent-based simulation on AgentSociety2 as a policy sandbox: six LLM-driven focal elders are embedded in a 20-person rules-layer support network, profiles are generated from CLHLS 2021 and CHARLS survey distributions, and the three interventions run in months 7–18 and are then withdrawn for six months. Methodologically, we contribute a transferable scientific discipline: preregistration and validity gates, root-cause classification of failures, LLM-call accounting including invalid runs, a ledger stating whether each result is fixed by rules or parameters, and dynamics calibration accepted on held-out targets. Three main results follow. (1) **Data anchoring works**: survey-based profiles beat ordinary random ones on input fidelity; after calibrating loneliness dynamics, the held-out three-year persistence error falls from 43.1 to 4.6 percentage points, and survey-anchored profiles beat random ones on output error (4.3 vs. 7.3 pp). (2) **Where relationships go after withdrawal**: split by the layers of the convoy model, social-worker visits activate family contact only while the service runs and leave none behind; the peer contact left after mutual-aid pairing is supplied entirely by the pair-persistence parameter, and LLM elders almost never contact their pair partners themselves; volunteer matching leaves no everyday relationship; all three services leave only a help-seeking channel to former providers, and removing old ties raises post-withdrawal isolation by +3.1 pp. Social-worker visits are strongest in-window (isolation 7.9% vs. a 29.4% baseline), and running them together with mutual-aid pairing is sub-additive. Each layer's post-withdrawal path is set by its parameter, so we state which quantities must be measured in practice. (3) **A cross-model comparison exposed an environment defect**: a more boundary-respecting model went largely silent on the old router, which turned out to phrase "cannot write in this call" as "cannot contact anyone"; after the fix it acts. Case results are projections under explicit parameters for generating testable practice hypotheses, not effect estimates for real populations; every number reproduces from released scripts.

Keywords: social work; older adults living alone; social support; mutual-aid pairing; agent-based simulation; scientific discipline

1 Introduction

Loneliness and social isolation among Chinese adults aged 65+ living alone are large in scale: in the CLHLS 2008/09 wave, 52% of those living alone reported feeling lonely, nearly twice the rate among those not living alone (29.5%)[1] (background only; our calibration uses the 2021 wave and the "often/always" definition, so the figures are not directly comparable). Communities respond through three main channels belonging to three policy fields: **social-worker visits** (social work: professional outreach by risk tier), **mutual-aid pairing** (social security: mutual-aid elder care in which healthy younger-old residents are paired with older-old ones and sustained by time-bank credits), and **volunteer matching** (social governance: volunteers dispatched on demand through a digital platform in community governance). The three are often combined in practice, yet three questions remain hard to answer empirically:

(1) Which relationships remain once a service is withdrawn? Every service ends. In the convoy model[2] the three services act on family (inner layer), peers (middle layer) and weak ties (outer layer); whether any gain survives withdrawal depends on whether that layer's own maintenance has been changed. Evaluations almost always measure outcomes during the programme or at its end[3], and the counterfactual trajectory after withdrawal cannot be observed, because the same elders cannot both receive and not receive an intervention.

(2) When is entry least costly? Waiting lists and budgets make start timing a real policy variable, yet its marginal cost has never been systematically quantified.

(3) Can such a simulation be trusted? Generative agents[4] produce reasoned decisions in a controlled environment and allow "parallel worlds" to be replayed under the same seed, which suits the first two

questions; but they face a threefold challenge: are the data fabricated, are the conclusions reproducible, and are the "findings" merely echoes of the environment's rules?

Using social support for older adults living alone as the case, we take the first question as the main line: the three services are not three parallel options but three projections of the same proposition on three relationship layers, and the method discipline of the third question is what makes the projection credible. Our contributions are threefold:

1. **A layer-by-layer projection of where relationships go after withdrawal, with a measurement agenda**: contacts after withdrawal are split by relationship layer, showing which quantity governs each layer's persistence and what should be measured in practice (Section 5.3).

2. **A transferable scientific discipline**: preregistered hypotheses and validity gates, archiving and root-cause classification of failed runs, LLM-call accounting including invalid runs, and a **result-source ledger** stating which results follow directly from rules and parameters (Table 7).

3. **Data anchoring with held-out validation**: profiles are generated from survey distributions, and a preregistered ablation shows they are closer to reality than ordinary random ones; loneliness dynamics are calibrated by the propose–evaluate loop on training targets and accepted on held-out targets (Section 7.2).

The layered policy sandbox (an LLM focal layer embedded in a rules layer that makes no LLM calls, with intervention mechanisms and parameters externalised) carries all three contributions (Section 5.4). Case results are scenario projections under explicit parameters, used to demonstrate the method and to generate testable practice hypotheses. Core conclusions and innovations are summarised in Section 9; reviewers' concerns and the corresponding sections are mapped in Appendix Table 10.

2 Related Work and Theoretical Basis

Social support theory and the convoy model. The convoy model[2] views an older person's social ties as layered convoys moving through the life course: the inner circle (spouse, children) is stable but eroded by widowhood and health shocks, while the outer circle (neighbours, friends) is more fluid. Our relationship ledger operationalises this structure directly: every tie has a strength, a type, and a most-recent-contact month, and decays when contact lapses.

The empirical shape of tie decay. Burt's cross-study fits show that decay rates are non-constant and follow a power function—"older ties are more stable"[5]; tie termination among older adults has also been measured directly[6]. In Section 7 we measure our model's survival curves against Burt's yardstick and report the structural deviation candidly.

Effect-size evidence for loneliness interventions. The meta-analysis by Masi et al. provides an effect-size yardstick for four intervention classes (social-cognitive interventions largest, $d = -0.598$)[7]; the subtype breakdown of Hoang et al. shows technology-based interventions with CIs crossing zero and multi-component interventions significant in community settings[8]. Two casework RCTs (older adults living alone[9]; community-dwelling frail elders[10]) point in the same direction with conservative ITT estimates. Quasi-experimental timebank evidence indicates effects lasting at least to programme end with no crowding-out[3]. This literature supplies the direction and relative strength of the three interventions; our intervention parameter values have not yet been aligned with these effect sizes (Section 3.4).

Generative-agent simulation. Park et al. established the paradigm of LLM agents producing believable social behaviour[4]. This paper differs in its policy-comparison purpose: a preregistered experimental design, calibration to real survey data, and systematic validity checking.

3 Method: Layered Hybrid Simulation

3.1 Overall architecture

The simulation is built on the AgentSociety2 framework (Ray parallelism, full tracing, replayable decisions). Twenty older adults living alone form a support network; six of them are **LLM focal elders** (PersonAgent, model gpt-5.6-sol) who, each month, run a multi-round tool loop to perceive the environment, retrieve memory, act, and articulate a natural-language rationale. The remaining fourteen residents and all network dynamics (tie decay, monthly events, intervention mechanics) are driven by the **rules layer** and consume no LLM calls. The focal layer supplies mechanistic and narrative evidence; the rules layer preserves the integrity of the network structure and supports zero-cost large-sample sensitivity scans. Measured call density is roughly 13.8 calls per focal elder per month. **The two layers generate contact differently**: rules-layer elders automatically receive contact each month according to their child-contact frequency and extroversion; focal elders receive contact only through their own tool actions (calling family, visiting neighbours, seeking help, attending activities) and through visits by intervention actors—a month in which the LLM takes no action is recorded as a month without contact. This difference is the direct cause of the threefold gap in isolation levels between layers reported in

Section 5.4, and it means the focal-layer isolation rate measures both "is the elder isolated" and "did the LLM act". Figure 1 shows the overall structure.

3.2 Environment router

Focal elders cannot call environment tools directly. They can only send a one-sentence natural-language request through `ask_env`, declaring whether it is read-only (observing) or write (acting). The code-generation router passes the sentence and the tool signatures available in that mode to another model, which writes call code that is executed in a sandbox; the result returns to the elder in natural language. Registered tools come in two kinds: read-only tools (observing one's own situation, the community report) and write tools (calling, visiting, asking for help, joining activities); read-only requests can only use the former. This layer is the only interface between model and environment, so how it states capability boundaries directly affects whether agents act. Before 2026-10-20, read-only answers did not distinguish "cannot write in this call" from "no such function"; after the fix, questions about whether a function exists are answered from the registry ("the tool exists; this read-only call cannot invoke it; use readonly=false to act"), and any generated answer still denying a registered tool is replaced and not cached (the incident is described in Section 7.7).

3.3 Environment and the relationship ledger

The environment module (ElderSupportEnv) maintains a relationship ledger for every elder: tie type (child / neighbour / friend / volunteer / social worker), strength $s \in [0, 1]$, monthly contact frequency, and most recent contact month. A tie with no contact in a given month decays as $s \leftarrow s \times (1 - \delta)$ (baseline $\delta = 0.06$; sensitivity scan 0.15/0.25) and is deleted at $s < 0.05$. Monthly events are of three kinds: illness and falls (both rising with age and lowering health) and a neighbour or friend moving away (deleting that tie), with baseline monthly rates of 0.10, 0.03 and 0.02; these rates are modelling assumptions (appendix parameter table). All simulated elders live alone and no spouse is modelled, so there is no widowhood event. Contact with children is not an event; it is generated each month from the contact frequency of each child tie. Loneliness $\ell \in [0, 1]$ relaxes each month by half towards a steady state set by the month's emotional contacts, tie count, health and extroversion (appendix parameter table); the formula has no event term, and events enter only indirectly through health. Loneliness is therefore a smoothed function of the isolation rate, not an independent outcome. The cut-off $\ell \geq 0.6$ is hand-set and not calibrated against the survey distribution, so it is not treated as an aligned target.

3.4 The three interventions

Selection logic. The three are not social work's three practice methods (casework, group work, community work), nor do they sit at the same level: social-worker visits are a professional service, mutual-aid pairing is a reciprocity mechanism carried by a time bank, and volunteer matching is volunteer service brokered by a digital platform. In what follows we refer to the three conditions by their code names casework, timebank and platform. What we compare is one dimension they share: which layer of an elder's support network they supplement, and what relational logic sustains it. Within the convoy model[2], social-worker outreach injects a formal support node and tries to repair inner-circle family ties, corresponding to formal support filling in where informal support is missing in the hierarchical compensatory model[11]; the timebank builds reciprocal peer ties in the middle circle, sustained by the norm of reciprocity[12]; the platform matches volunteers on demand in the outer circle, supplying weak ties[13]. The three logics correspond to "who fills in", "can it sustain itself" and "can it reach on demand", which is why they suit a comparison of whether ties persist after withdrawal. In policy terms the three belong to social work (professional outreach), social security (mutual-aid elder care organised through credits) and social governance (volunteer platforms in community governance), covering the three main channels through which communities support older adults living alone. In practice the three are often combined; we compare them separately to identify mechanisms, not because they are mutually exclusive or interchangeable. Table 1 lists, for each intervention, the need addressed, the service actions as implemented in code, the relational mechanism, the post-withdrawal condition and the degree of implementation.

- **casework — social-worker visits (risk-targeted outreach)**: after risk ranking, social workers visit high-risk elders and reconnect them with family; top-down and precise, but constrained by caseload. Only three components of case management are implemented—risk screening, regular visits and family linking—with no needs assessment, service plan, follow-up or case closure, so we do not call it case management. The direction of effect follows the Masi meta-analysis[7] and the two casework RCTs[9, 10]; the coverage cap equals the number of workers × caseload.
- **timebank — mutual-aid pairing (time bank)**: younger-old healthy elders are paired with high-risk elders and exchange services for credits under automatic rules; the pairings are real interpersonal ties and persist after withdrawal with probability post_persist (midpoint 0.5; rules-layer scan 0–1 including a zero control, Table 3). The mechanism follows the Hong Kong timebank quasi-experiment[3]; no literature point estimate

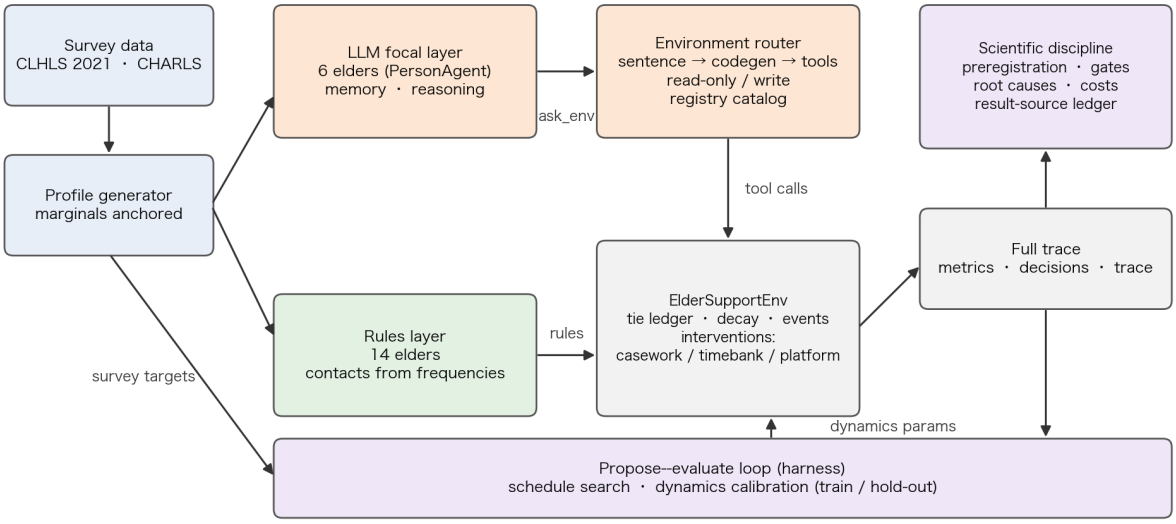


Figure 1: System architecture. Profiles are generated from survey marginals; the six LLM focal elders can reach the environment only through one-sentence requests to the router, while the 14 rules-layer elders make contacts from their frequencies; the environment records everything, and the scientific discipline and the propose–evaluate loop act on the trace and on dynamics parameters.

exists for post-withdrawal persistence, so it is treated as a scan.

- **platform — volunteer matching (digital platform)**: responsive, broad-coverage, rotating volunteers, shallow ties. Its effect is set weakest of the three, with the direction following meta-analytic evidence that technology-based interventions have CIs crossing zero[8].

Nature of the parameters. The literature supplies only the direction and relative strength of the three interventions; their specific probabilities (e.g. 0.8 that a social-worker visit counts as emotional contact, 0.5 for a family link, 0.85 that a timebank visit takes place in a month, a 0.35 cap on platform-volunteer tie strength) are modelling choices, not converted from published effect sizes; values are listed in the appendix parameter table. The ratio of the three interventions' effect sizes should therefore not be read as an estimate of real-world service effects.

Unequal inputs. The three interventions' service volumes are not equalised: social-worker outreach covers a fixed workers × caseload = 16 elders each month (80% of the 20); the timebank is limited by the number of available helpers; the platform dispatches only to elders with a life event that month or two consecutive months without emotional contact, at 60% coverage. Differences in outcomes between interventions therefore combine "how much input" with "the mechanism itself" and cannot be read directly as one mechanism being better than another.

3.5 Data calibration

The profile generator is anchored dimension by dimension to real surveys: age band, sex (63.8% female), self-rated health, a five-level loneliness baseline, number of surviving children, and the share of children who visit regularly (78.7%) come from the CLHLS 2021 living-alone subsample ($n = 1707$); child proximity comes from CHARLS 2015. Extroversion and the number and strength of neighbour and friend ties have no survey anchor and are modelling assumptions. Longitudinal targets (three-year loneliness onset 27.2% and persistence 53.3% among those living alone) were also extracted from Harmonized CHARLS 2011–2018; they are used for the dynamics calibration and held-out check in Section 7.2. Main-batch profiles were verified to be byte-identical to profiles regenerated from the survey distributions (seeds 1–3). **Known bias**: CLHLS oversamples the oldest-old, and age bands are sampled here from the unweighted distribution (69.4% aged 80+), so the profiles skew old; reweighting with the CHARLS weighted age distribution is left for future work. Outcome-definition mappings are documented in the appendix and repository. **Scope of the calibration**: what is anchored is only the *marginal distribution* of all 20 residents' profiles, not the contact process. Rules-layer elders' monthly contact is generated by bands of child proximity and regular-visit status, but the monthly contact count for each band is a modelling assumption, not calibrated to survey data; focal elders' contact is generated by LLM behaviour (Section 3.1), and their baseline isolation rate (56.9%) far exceeds that of the rules layer (17.7%). Strictly, therefore, only the marginal profile distribution of the 20 residents is aligned with the surveys; neither the

Table 1: Need, action and mechanism of the three interventions (the action column follows the actual implementation in `custom/envs/elder_support_env.py`).

Condition	Need / barrier	Service actions (as implemented)	Relational mechanism	After withdrawal	Implementation
casework	High-risk elders lack emotional support; family contact has thinned	Top 16 by risk score visited once a month (emotional contact with probability 0.8); with probability 0.5, contact with the strongest child tie	Injected formal node; activates inner circle	Visits stop; social-worker tie fades at the decay rate; no exit plan	Simplified: no needs assessment, service plan, follow-up or closure
timebank	Lack of peer companionship; healthy younger-old lack a social role	Elders with health ≥ 0.6 and age ≤ 78 paired with high-risk elders (at most 2 each), visit with probability 0.85 a month; both ties strengthen, helper earns credits	Peer reciprocity; middle circle	Pairs kept; interact each month with probability post_persist	Simplified: credits cannot be redeemed; trust formation and organisational upkeep not modelled
platform	No one to call on when a life event occurs	Elders with an event that month or two months without emotional contact receive a random volunteer at 0.6 coverage; tie strength capped at 0.35	On-demand matching; outer-circle weak ties	No more dispatches; volunteer ties fade at the decay rate	Simplified: digital skills and access barriers not modelled; dispatch triggered by the environment, not requested by the elder

All three interventions share the same outcome measures: isolation rate, loneliness, and focal elders' help-request counts (descriptive).

rules-layer contact frequencies nor the LLM-layer contact dynamics are calibrated. There are two output-side external checks: the tie-survival curve does not match the Burt fit (Section 7); and after calibration of loneliness dynamics in the rules layer, the held-out three-year persistence error is 4.6 pp (Section 7.2). The latter is a post-hoc calibration; the main batch ran on the default, uncalibrated dynamics, so the numbers in Section 5 are unaffected. **Profile source of rules-layer scans.** The timing and dose scan, the harness loop, and the decay-rate and post_persist scans in Section 6 originally used placeholder profiles (fixed assumed parameters for every dimension). On 2026-10-21 the timing, decay and post_persist scans were rerun with calibrated profiles and keep the same directions (`tests/*_calibrated.json`); the tables report the original runs, and the comparison is in Section 7.2.

3.6 Preregistration and validity gates

Hypotheses (H1–H4), primary and secondary outcomes, seeds, and validity criteria were written into script headers before the formal batch was launched: **primary outcome** isolation_rate (monthly isolation rate, defined as the share of elders with no emotional-support contact in the month); **secondary outcome** avg_loneliness; deep_isolation_rate enters sensitivity analyses only, and avg_active_ties is descriptive only. **Validity gates**: a run must complete 24 months, have non-empty decision records, focal-elder silence rate $\leq$10%, and LLM call error rate $\leq$5%; any violation renders the run invalid and triggers a full rerun, with invalid runs archived rather than deleted. With $n = 3$ seeds we perform no significance tests and claim no stable ranking; H4 is a within-run paired contrast (post-withdrawal − intervention window) and is flagged as exploratory.

Nature and limits of the primary outcome. isolation_rate is a monthly snapshot of "was there any emotional contact this month". Any presence-based service delivered monthly (casework triggers one emotional contact with probability 0.8 for each covered elder each month) drives it to zero by definition, so within the intervention window the indicator mainly measures "is the service present"; the step changes at months 7 and 19 in Fig. 2 arise from this. Loneliness ℓ is a continuous quantity with homeostatic dynamics (each month it moves halfway toward a set point determined by contact volume, tie count, health, and personality) and is closer to a genuine outcome. The preregistration lists it as secondary; we do not alter that status, but we **report the two side by side** and, when interpreting effect sizes, treat loneliness as the outcome and the isolation rate as the "presence" indicator.

4 Experimental Design

Four conditions (none / casework / timebank / platform) $\times$ 3 seeds $\times$ 24 months, targeting 12 valid runs, of which 11 were frozen: platform seed 2 was ruled invalid because the focal elders' silence rate reached 11.3% (17/150 agent-months), above the 10% gate. A 2026-10-20 trace review shows all 150 agent-steps completed, the LLM error rate was 0, and the 17 silent months

Table 2: Primary outcome by window (isolation rate, %; condition means). Rebound = post-withdrawal − intervention window, pp.

Condition	n	Interv.	Post	Rebound
none	3	29.4	27.2	-2.2
casework	3	**7.9**	27.5	**+19.6**
timebank	3	24.7	25.6	+0.8
platform	2	23.5	27.5	+4.0

were spread across five elders rather than a gateway-timeout hole. A same-configuration rerun on 2026-10-19 reached 60.7% silence (91/150) with an LLM error rate of 0; the trace shows elders asking the environment whether they could contact others, the read-only router answering "no", and the elders then ending the step. This is a routing design problem (Section 7.7), not a model trait. That rerun is also invalid and is not folded into the main results, so the platform arm is reported with $n = 2$; the intervention window spans months 7–18, with post-withdrawal observation in months 19–24. Hypotheses (quoted verbatim from the preregistration, where "casework" was described as case management; the name was narrowed to match the actual implementation after the formal batch, see Section 3.4): **H1** all three interventions lower the isolation rate during the window, with casework lowering it most; **H2** loneliness improves in the same direction during the window; **H3** the three interventions leave distinguishable behavioural signatures (help-seeking routes, pairing, volunteer matching); **H4** (exploratory) casework rebounds more than the timebank after withdrawal—structural mutual aid is more sustainable (preregistered wording reproduced verbatim; Section 5 explains that this pattern is fixed by explicit mechanism assumptions and is not read as a finding. The "rebound" mixes in differences in in-window level—a weak intervention has little room to rebound—so we compare sustainability by post-withdrawal level and its residual relative to the never-treated arm; rebound is listed in tables only).

5 Main Results

5.1 In-window effects (H1, H2)

During the window, casework yields an isolation rate of 7.9%, a relative reduction of 73% against the 29.4% baseline; the same-seed paired difference is -21.5pp (paired n =3). The timebank (24.7%) and platform (23.5%) improve outcomes only modestly. Loneliness (reported alongside) moves in the same direction but with a much smaller magnitude: casework 29.4% vs. baseline 37.7% (a relative reduction of about 22%, against 73% for the isolation rate)—the gap between the two indicators is itself evidence that the isolation rate mainly measures service presence. This ordering follows the presence effects set by the parameters: a visit counts as emotional contact with probability 0.8 for the social worker, 0.6 for timebank recipients and 0.3 for the platform, and the platform only dispatches after an event or two months without contact; loneliness falls with emotional contact through the steady-state formula. H1/H2 are therefore fixed by parameters in this model and do not constitute independent support.

5.2 Post-withdrawal sustainability (H4)

After withdrawal, the casework isolation rate rebounds by +19.6pp (same-seed paired difference +21.8pp), all but erasing the in-window advantage; the timebank rebounds by +0.8pp (Figs. 2 and 3). A 20-seed rules-layer scan shows the same shape (casework rebound +7.9 to +9.5pp, timebank +2.1pp).

Result source. Both mechanisms are written into the environment. Casework's emotional contact is triggered directly by the social worker's monthly visit and disappears on withdrawal; timebank pairings continue to interact after withdrawal with probability post_persist (midpoint 0.5), and the code comment explicitly labels this persistence probability "the key mechanism assumption for H4". LLM focal elders and rules-layer elders inhabit the same environment and share the same indicator definition, so the rules layer and the LLM layer are *not* two independent lines of evidence, and agreement between them does not constitute cross-layer corroboration. The accurate statement of H4 is: **under explicitly encoded mechanism assumptions, the simulation reproduces the expected sustainability divergence and provides its sensitivity range** (caseload 4–12 does not alter the rebound; the post_persist scan follows). Its value lies in turning the practice intuition "dependency-type support vanishes on exit, reciprocal ties partly persist" into a parameterised, refutable model—not in proving it true.

The post_persist = 0 control (rules layer, 20 seeds). To test directly whether the divergence is manufactured by this parameter, we scanned post_persist $\in \{0, 0.3, 0.5, 0.7, 1.0\}$ in the rules layer (Table 3). The result has two halves. (1) **Post-withdrawal level**: at post_persist = 0 the timebank's post-withdrawal isolation rate is 12.0%, identical to casework (12.0%)—*once pairings do not persist, the timebank has no sustainability advantage whatsoever*; as post_persist rises to 0.5/0.7 the post-withdrawal level falls to 8.5%/8.3%, so the advantage is supplied entirely by the parameter. (2) **Rebound magnitude**: even at post_persist = 0 the timebank rebounds less (+5.4pp) than casework (+9.1pp; 17/20 seeds), but this is arithmetic: the timebank's in-window effect is weaker (6.6% vs. 2.9%), leaving less room to rebound, and both return to the same level. **Comparing sustainability by rebound magnitude therefore systematically**

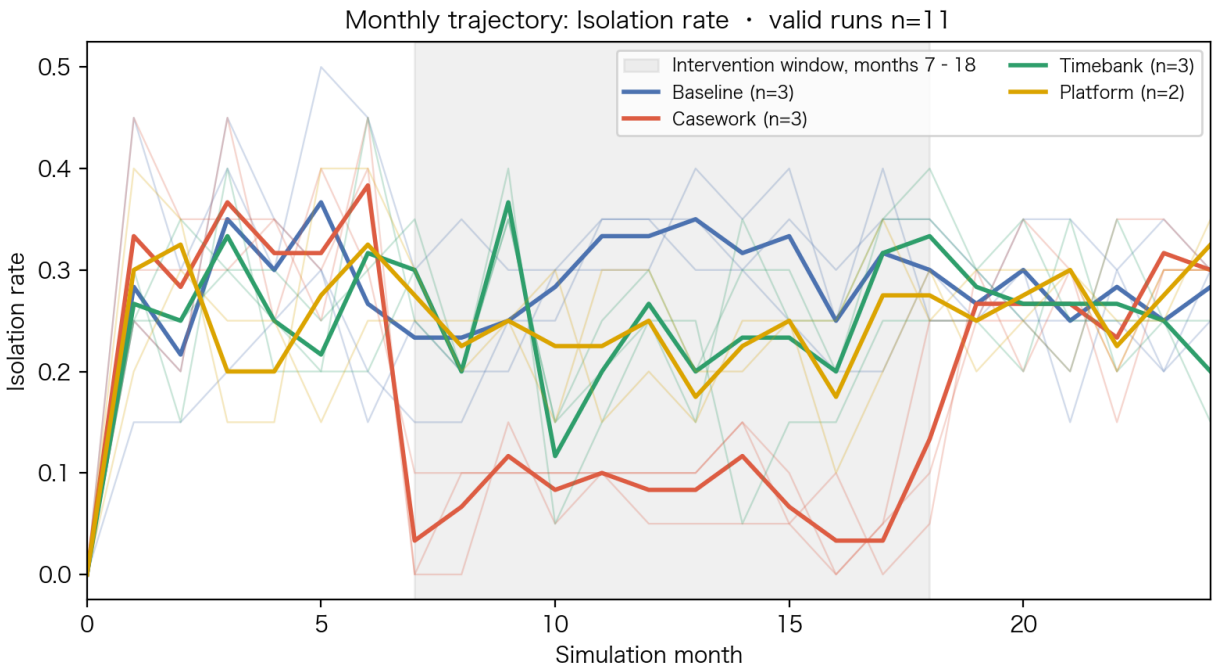


Figure 2: Monthly isolation-rate trajectories (thick lines: condition means; thin lines: individual seeds; shading: intervention window, months 7–18). The casework arm shows step changes at months 7 and 19, corresponding to the entry and exit of a presence-based service (see Section 3.6 on the nature of the primary outcome).

Table 3: Rules-layer post_persist scan (20 seeds; all elders in the rules layer, so absolute levels are not comparable with the LLM batch). Isolation rate as window mean (%); rebound = post − in-window (pp); last column: seeds in which the timebank rebound is smaller than same-seed casework.

Config	In-win.	Post	Reb.	Reb.<cw
none	—	14.5	—	—
casework	2.9	12.0	+9.1	—
timebank, pp= 0	6.6	12.0	+5.4	17/20
timebank, pp= 0.3	6.6	10.5	+3.9	18/20
timebank, pp= 0.5	6.6	8.5	+1.9	19/20
timebank, pp= 0.7	6.6	8.3	+1.7	19/20
timebank, pp= 1	6.6	5.4	-1.2	20/20

favours weak interventions; the correct quantity is the post-withdrawal level or the residual effect relative to baseline, and our main table (Table 2) reports both. A side observation: both interventions' post-withdrawal levels (12.0%) lie below never-intervened (14.5%), reflecting slowly decaying residual tie strength built during the window—again a direct consequence of δ.

5.3 Where relationships go after withdrawal: a split by relationship layer

Here "layer" means the relationship layers of the convoy model (family, peers, weak ties), not the focal and rules layers of Section 5.4. We split the emotional contact each elder receives per month by the type of each tie in the relationship ledger (rules layer, 20 seeds, observed to month 30, `tests/scan_layers.py`), and split whom the LLM elders contact on their own initiative (valid main-batch runs, `tools/llm_layer_actions.py`); both analyses were preregistered first and make no LLM calls (Table 4).

Inner layer: family activation does not last. While casework runs, elders receive +0.176 more emotional contacts per month through family than without intervention (95% interval [+0.129, +0.220]), all of it from the social worker's "family link" step (without that step the level equals no intervention); after withdrawal the difference is +0.010 ([-0.051, +0.069], crossing zero; -0.002 with placeholder profiles). The LLM layer agrees: social workers link focal elders to family 0.41 times per month in-window, and elders' own contacts with family do not rise after withdrawal (0.046 vs. 0.028 with no intervention). Whether social workers "crowd out" children's contact cannot be tested, because no rule in the model represents it.

Middle layer: pair persistence is supplied by a parameter, and LLM elders do not maintain pairs. After timebank withdrawal elders still receive 0.182 emotional contacts per month through pair partners, falling to 0.061 when persistence is zeroed; contact with neighbours and friends is unaffected (-0.003). But LLM elders almost never contact their pair partners themselves (0.009 per month in-window, 0.000 after withdrawal): the "reciprocal persistence" assumed in the rules layer has no counterpart in active LLM

behaviour. Whether reciprocity can run on its own without the credit system is likewise not modelled.

Outer layer: weak ties do not settle. After volunteer matching ends, each elder still has 1.54 volunteer ties on the ledger, but receives only 0.067 emotional contacts per month through them. The model has no mechanism for weak ties to become familiar ones, so the conversion rate cannot be tested.

A residue common to all three layers: the help-seeking channel. The preregistration predicted zero post-withdrawal contact through the layer for the zeroed-persistence and volunteer arms; we observed 0.061 and 0.067, and casework leaves 0.183 per month through the former social worker. Old ties remain on the ledger at withdrawal, and elders facing an event seek help from their strongest ties. A control (`tests/scan_layers_cleanup.py`, preregistered first) removes the old service ties just before withdrawal: these contacts disappear entirely, and post-withdrawal isolation rises by +3.1, +2.2 and +1.7 pp for casework, volunteer matching and zeroed persistence (casework [+2.1, +4.0]); casework's remaining post-withdrawal advantage over no intervention thus comes almost entirely from this channel. The LLM layer points the same way: after withdrawal elders still contact former social workers (0.102 per month) and volunteers (0.111) only through help requests, never by calling or visiting. This follows from the help-seeking rule and the leftover ties, not from relationships persisting on their own; whether, and how much, real elders keep asking former providers for help after a service ends, and whether providers respond, must be measured in the field.

5.4 Layer decomposition

Table 5 decomposes the window means of the six focal elders and the fourteen rules-layer elders from the replays. Three observations. (1) **Focal elders' baseline isolation rate is about three times that of rules-layer elders** (none: 56.9% vs. 17.7%), a consequence of the contact-generation difference described in Section 3.1—an LLM elder that does not act has no contact. (2) **The LLM layer dominates the macro effect**: focal elders are 30% of the population but contribute 58% of the baseline isolation rate, and casework's paired effect on the focal layer (in-window -44.0pp, rebound +55.1pp) is three to seven times its effect on the rules layer (-11.9pp, +7.5pp). "Generative-agent simulation" is therefore not decorative—but it also means the headline results are highly sensitive to LLM behavioural traits, including silence. (3) **The timebank's "no rebound" is mainly a rules-layer contribution**: the rules-layer paired rebound is -0.8pp (post_persist at work), whereas the raw LLM-layer rebound is +12.0pp, comparable to the platform arm (+13.9pp) and far below casework (+55.1pp). What the LLM layer supports is thus "casework rebounds far more than the other two", not "the timebank barely rebounds". Loneliness moves in the same direction in both layers, with gentler magnitudes.

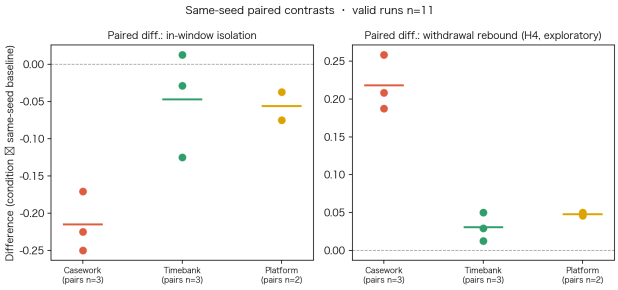


Figure 3: Same-seed paired contrasts (left: in-window effect; right: post-withdrawal rebound, exploratory).

Do the focal-layer numbers measure isolation or silence? The "excluding silent months" columns of Table 5 count only focal agent-months with at least one decision record (silence rates in valid runs: 4.0–10.0%) and answer this question. After exclusion, focal isolation rates fall by 1–8 pp in every condition, the ranking is unchanged, and the LLM layer's dominance is unchanged; but the timebank's LLM-layer rebound shrinks from +12.0pp to +4.2pp—most of it came from silent months in the post-withdrawal window—and after exclusion it sits between the rules layer (-0.8pp) and casework (+48.8pp).

5.5 Help seeking (H3)

Focal elders' help-request counts are given in the footnote[1]. Given the target, success is decided by $0.15 + 0.8\,s\times$reachability, so differences in success rate mainly reflect whom requests go to; with three seeds these are descriptive only.

5.6 Cross-layer discrepancy

The in-window ranking is not fully consistent across layers: the rules layer puts timebank above platform, while the LLM layer puts platform slightly above timebank (platform has only 2 valid seeds; the difference cannot be separated from seed noise). It is mechanistically explicable: the LLM layer's platform intervention introduces new volunteer ties and raises active-tie counts. We report this as "a cross-layer difference and its explanation" rather than concealing it—exposing exactly such inconsistencies is where the layered design earns its keep.

[1] Over the 11 valid runs (`tests/behavior_signature.json`): none, 3 runs, 249 requests / 183 successful (73.5%); timebank 247/185 (74.9%); casework 412/271 (65.8%); platform, 2 runs, 159/94 (59.1%).

Table 4: Where relationships go after withdrawal. A: rules layer, emotional contacts per elder per month through each layer (20 seeds, calibrated profiles), in-window months 7–18 / post-withdrawal months 19–30. B: LLM main batch, focal elders' self-initiated contacts per month with each layer (old router, valid runs), in-window / post-withdrawal months 19–24.

	Family (inner)	Peers (middle)	Pair (middle)	Volunteer (outer)	Social worker
A. Rules layer					
None	3.610 / 3.608	0.140 / 0.105	–	–	–
Casework	3.786 / 3.618	0.147 / 0.094	–	–	0.808 / 0.183
Casework (no family link)	3.595 / 3.614	0.140 / 0.102	–	–	0.812 / 0.182
Timebank	3.641 / 3.596	0.143 / 0.102	0.257 / 0.182	–	–
Timebank (persistence zeroed)	3.641 / 3.632	0.143 / 0.101	0.257 / 0.061	–	–
Platform	3.591 / 3.633	0.135 / 0.099	–	0.089 / 0.067	–
B. LLM main batch (n = valid runs)					
None ($n = 3$)	0.060 / 0.028	1.139 / 1.231	–	–	–
Casework ($n = 3$)	0.060 / 0.046	0.917 / 0.954	–	–	0.093 / 0.102
Timebank ($n = 3$)	0.046 / 0.019	1.148 / 1.028	0.009 / 0.000	–	–
Platform ($n = 2$)	0.035 / 0.000	0.986 / 1.042	–	0.090 / 0.111	–

Rules-layer family contact comes from children's fixed monthly contact, so it is far higher than the contacts LLM elders initiate; compare only within each panel. LLM help requests do not record whether a deep talk occurred, so "only through help requests" means no calls or visits.

Table 5: Layer decomposition (window means; isolation in %, loneliness 0–1). Focal = six LLM elders, Rules = fourteen rules-layer elders; "excl. silent" = only focal agent-months with a decision record; paired difference = condition − none under the same seed.

	Focal isolation		Focal, excl. silent		Rules isolation		Loneliness focal/rules	
Condition	Window	Post	Window	Post	Window	Post	Window	Post
none	56.9	51.8	53.2	50.9	17.7	16.7	0.56/0.30	0.58/0.33
casework	13.0	63.0	12.0	58.5	5.8	12.3	0.42/0.24	0.52/0.29
timebank	45.8	52.8	43.0	44.9	15.7	13.9	0.54/0.29	0.57/0.31
platform	43.1	56.9	41.2	54.4	15.2	14.9	0.52/0.28	0.53/0.29
Paired difference vs. none (pp): in-window / rebound								
casework	-44.0 / +55.1		-41.2 / +48.8		-11.9 / +7.5			
timebank	-11.1 / +12.0		-10.2 / +4.2		-2.0 / -0.8			
platform	-11.1 / +13.9		-7.8 / +9.5		-3.3 / +0.9			

6 Mechanism Scans: Timing, Dosage, Combination and the Propose–Evaluate Loop

6.1 Entry timing

A preregistered rules-layer scan (start month $\{3, 5, 7, 9, 12\}\times$ duration $\{6, 12, 18\}\times$ 3 conditions, 20 seeds; baseline full-horizon isolation rate 15.0%) yields three findings:

1. **Interventions act immediately**: within the window, the isolation rate barely varies with start month—the entire cost of delay comes from unprotected waiting months, not from "missing a critical window";

2. **A directional estimate of delay cost**: each month of postponement raises the two-year mean isolation rate by about +0.15pp (casework) / +0.18pp (timebank), about 3.5 pp (casework) over 24 months against a 15.0% baseline; the platform's total effect is weak, so it is timing-insensitive (+0.00pp). The magnitude is fixed entirely by the rule property "interventions act immediately" (Table 7) and is offered only as a directional estimate under the rules-layer mechanism;

3. **Rebound is dose-independent**: extending casework from 6 to 12 months does not soften the post-withdrawal rebound (+7.9 to +9.5pp; the independent post_persist scan measures +9.1pp, so the two scans corroborate each other)—the rebound of dependency-type support is a property of the mechanism, not a symptom of underdosing.

Disclosures: a later start implies a shorter post-withdrawal observation window (a confound), and the 18-month duration has no post-window at all; rules-layer absolute levels are not compared with the LLM batch—only direction and magnitude are used. An LLM-behavioural-layer timing contrast across prevention / early-warning / crisis stages has *not* been run; it is future work, and we do not write plans as results.

6.2 Propose–evaluate loop (harness)

We demonstrate an automated research loop under preregistration constraints: an LLM designer proposes candidate configurations round by round within a preregistered search space (condition × start month 3–12 × duration 6–12, objective: minimise full-horizon isolation rate, resource cap duration ≤12), a deterministic rules-layer evaluator executes them, and all 4 rounds are logged to disk. The loop converges to casework start=4 dur=12 (full-horizon isolation rate 7.70%), far within one seed standard deviation of the best grid point under the resource cap (start month 3, 12 months: 7.72% ± 2.29); a rerun on 2026-10-22

under the original preregistration found the same optimum, as did a rerun with calibrated profiles (8.08%, `tests/harness_loop_result_calibrated.json`)—**the loop works as a procedure, and we report candidly that it rediscovered the grid's plateau without surpassing it**. The guardrails (LLM proposes only; evaluation is deterministic; space and round count preregistered; every round logged) ensure the demonstration introduces no selective reporting. In addition, the per-run automated analysis–reflection mechanism (analyze.py) currently implements "detect and flag": any change to the experimental configuration requires human review, preventing mid-experiment design drift. Note that the loop only searches intervention schedules under fixed environment rules: it does not modify model parameters, nor does it carry lessons from one round into the next automatically. It shows that the loop runs and that selective reporting is blocked, not that the tool improves itself. Section 7.2 switches the proposal target to dynamics parameters with a held-out acceptance check, the next stage of this loop.

6.3 Combining interventions and hand-over

Section 5 compares the three interventions one at a time and leaves two practice questions open: does running them together add more than the sum of the parts, and which order lasts longer after withdrawal? We ran a preregistered rules-layer combination experiment (`tests/scan_combo.py`, committed before running), restricted to social-worker visits and mutual-aid pairing—the two services in the "social workers open the way, mutual-aid pairs carry on" idea of Section 10. Volunteer matching is not combined.

Design. Six arms, interventions within months 7–18 and observation to month 30 (twelve post-withdrawal months): A none; B casework 7–18; C timebank 7–18; E both at once 7–18, forming a 2×2 factorial with A–C; D casework 7–12 then timebank 13–18; F timebank 7–12 then casework 13–18, the same input as D in reverse order. "Hand-over" is an explicit parameter b (handover_boost): if the recipient had already been visited by a social worker when a pair forms, its post-withdrawal persistence probability rises from p to $p + b(1 - p)$; $b = 0$ is the zeroed control, and we scan $b \in \{0, 0.25, 0.5, 1\}$. Twenty seeds, calibrated profiles as the main analysis and placeholder profiles as sensitivity, with same-seed paired differences and bootstrap 95% intervals. The new combination type in the environment is off by default; regression tests confirm that single-intervention results are bit-identical.

Results. (1) **Running both at once is sub-additive**: the interaction $I = (E-A)-[(B-A)+(C-A)]$ is +5.6 pp in-window (95% interval [+3.6, +7.7]; lower isolation is better, so a positive value means the joint gain is smaller than the sum), +0.28 on the logit scale ([+0.05, +0.52]), and +6.5 pp with placeholder profiles; after withdrawal it is +1.1 pp ([-0.1, +2.4], crossing zero). Both interventions target the highest-risk elders first, isolation only records whether an elder had any emotional contact that month, so repeated contact with the same elder adds nothing, and casework alone already pushes in-window isolation down to 2.9%, close to the floor. The preregistration text gave the sign of sub-additivity as "< 0"; the formula and the results are unchanged, and we disclose the slip here. (2) **Order alone does not decide persistence**: at $b = 0$ the post-withdrawal D−F difference is +0.5 pp ([-0.5, +1.7], crossing zero), so the preregistered prediction "D below F" is not supported. (3) **The "carry on" advantage rests on an assumption**: at $b = 0$ the post-withdrawal D−C difference is -1.3 pp with calibrated profiles ([-2.8, -0.1], upper bound near zero) and -0.1 pp with placeholder profiles ([-1.4, +1.3], crossing zero), which needs $b \geq 0.25$ to fall reliably below C; at $b = 1$ it is -2.9 pp. The small advantage of D at $b = 0$ comes from pairs formed in months 13–18 being fresher at withdrawal, plus six earlier months of casework, not from a hand-over mechanism.

LLM layer. We added arms D and E on the fixed router with gpt-6-astra seed 1 (2,849 calls; silence 2.7% and 0.7%), reusing the same-router A–C runs (Section 7.7). With one seed this is descriptive only: the in-window interaction is +3.3 pp for isolation and +0.034 for loneliness, the same direction as the rules layer; post-withdrawal isolation under D is +4.2 pp above C (loneliness -0.013), with no carry-on advantage. The LLM elders showed no pair-maintaining behaviour after visits that the rules layer cannot explain, so, as planned, we did not add seeds.

Schedule loop. We extended the loop of Section 6.2 to combination and order (`tools/harness_loop_combo.py`, preregistered first): the designer proposes schedules within "at most twelve service-months in total, $b = 0$", 6 rounds used 6 calls, and the rules layer evaluates them. The best schedule is timebank in months 3–4 then casework in months 5–14, with mean isolation 7.9% over months 1–30, below the capped grid optimum (arm B, 9.8%). Most of the gain comes from starting earlier (round 1, casework alone from month 3, already reaches 9.2%), the rest from the persistence of the short early pairs after withdrawal; both are existing rules. As in Section 6.2, the loop found a schedule outside the grid but no new scientific finding.

7 Scientific Checks

7.1 Calibration alignment and result sources

Input distributions are aligned with the survey data dimension by dimension (Section 3.5); the only quantitative check on run-level dynamics is the comparison of the tie-survival curve with the Burt fit below, which fails; loneliness outcomes have no check yet (see Sec-

Table 6: Combined interventions (rules layer: 20 seeds, calibrated profiles, $b = 0$; LLM layer: fixed-router gpt-6-astra seed 1, single run). Isolation is the window mean (%), loneliness the post-withdrawal mean; post-withdrawal is months 19–30 in the rules layer and 19–24 in the LLM layer.

		Rules layer			LLM layer	
Arm	Schedule	Interv.	Post	Lonel.	Interv.	Post
A	none	16.1	16.4	0.332	26.2	25.8
B	casework 7–18	2.9	13.4	0.274	7.5	22.5
C	timebank 7–18	9.8	12.3	0.301	23.3	23.3
D	casework 7–12 → timebank 13–18	6.2	11.0	0.268	13.8	27.5
E	both 7–18	2.3	10.5	0.255	7.9	30.0
F	timebank 7–12 → casework 13–18	6.1	10.5	0.264	–	–

tion 3.5 for why). Table 7 states, item by item, what each result in this paper is determined by, answering the objection that LLM simulation merely echoes its rules: the main macro patterns are set by rules and parameters, and we report them as such.

7.2 Profile ablation and dynamics calibration

To answer "what do survey-based profiles buy over ordinary random ones?", we ran a preregistered profile ablation in the rules layer (`tests/scan_profile_ablation.py`; design and criteria committed in the script header before running): three profile sources, 20 seeds each, 20 elders per seed—*calibrated* (CLHLS marginals sampled dimension by dimension, as in the main batch), *uniform random* (anchored dimensions drawn uniformly over the same ranges), and *record resampling* (whole CLHLS respondents resampled, preserving cross-dimension correlation); non-anchored dimensions are identical across sources. Input fidelity is the five-dimension total variation distance; on the output side the no-intervention condition runs 48 months (12-month burn-in), and the primary metric is the mean absolute error over four targets: cross-sectional loneliness shares (15%, 40%), three-year onset (27.2%) and persistence (53.3%).

The results come in two steps (Table 8). **Before calibration**, calibrated and resampled profiles clearly beat uniform ones on input fidelity but not on output: all three sources show about 98% three-year persistence, i.e. loneliness states hardly change on their own, so the error comes from dynamics rather than profiles. We therefore added a monthly individual shock parameter (default 0; past results bit-identical) and fitted it with the propose–evaluate loop over 72 preregistered candidates (`tools/calibrate_dynamics.py`), using the first three targets for training and persistence as the held-out target, on seeds 101–120. Training error fell from 12.7 to 4.8 pp and **held-out error from 43.1 to 4.6 pp**; the Spearman correlation between training and held-out error across candidates is 0.46, so both preregistered acceptance checks pass. The selected point lies on the grid boundary, and by preregistration the grid was not extended afterwards. **After calibration**, rerunning the ablation on seeds 1–20, the two survey-anchored sources beat uniform random on output (4.3 and 4.2 vs. 7.3 pp). The dynamics were fitted only on calibrated profiles, which favours that source; but the resampled source, not used in fitting, reaches the same error, so the advantage is not only a fitting artefact. Across all sources, before and after calibration, the in-window ranking is casework > timebank > platform, with magnitudes that vary by source (about 1.5 times larger under uniform random): the direction is robust to the profile source, the magnitude depends on inputs.

Table 7: Result-source ledger: what each result in this paper is determined by.

Result	Nature	Source
Immediate response of isolation to intervention (step at month 7)	prescribed	indicator definition + monthly visit triggers contact
Casework rebound after withdrawal	prescribed	no visit, no contact
Timebank persistence after withdrawal (rules layer)	prescribed	post_persist param.; at zero the post-withdrawal level equals casework (Table 3)
Linear accrual of delay cost by month	prescribed	direct consequence of immediate onset
Tie-count paradox (ties remain, activation gone)	prescribed	decay slower than the monthly snapshot
Threefold isolation gap between layers	semi-prescribed	different contact generation (Sec. 3.1); magnitude not preset
Help success given the target	prescribed	$0.15 + 0.8\,s\times$reachability
Focal-layer isolation levels and request counts by condition	LLM-layer description	three seeds, descriptive only
Sub-additivity of casework and timebank run together	prescribed	same high-risk elders covered twice + floor of the monthly snapshot (Sec. 6.3)
Sequential casework→timebank beats timebank alone after withdrawal	parametric	hand-over parameter b and profile source; interval upper bound near zero at $b = 0$
Best schedule from the combination loop	prescribed	earlier start + pair persistence (post_persist)
Family activation does not last after withdrawal	prescribed	family links occur only in months with social-worker visits (Sec. 5.3)
Post-withdrawal help contacts with former providers	prescribed	old ties left on the ledger + help sought by tie strength; vanish when removed
LLM elders do not contact pair partners	LLM-layer description	valid main-batch runs, descriptive only

7.3 Structural limits of tie decay

Measuring our model's tie-survival curves against Burt's yardstick (rules layer, 50 persons × 48 months) reveals two **mechanism-level limits**: (1) tie loss is markedly slower than in the literature (4-year sur-

Table 8: Profile ablation (rules layer, 20 seeds × 20 elders). Lower TVD means closer to survey marginals; output error is the mean absolute error over four targets (pp).

	Calibrated	Uniform	Resampled
Input TVD	0.176	0.249	0.162
Output error (before)	19.6	16.0	21.8
Output error (after)	4.3	7.3	4.2
3-yr persistence (before / after)	98% / 59%	98% / 56%	100% / 49%

vival 78.8% vs. 38.3% under Burt's fit); (2) there is a survival floor: scanning δ from 0.06 to 0.4, 4-year survival bottoms out at 46.3% (at $\delta = 0.4$ it is already 47.8% after year 1 and flat thereafter), still above Burt's 38.3%—children with fixed contact schedules and high-frequency neighbours refresh their contact timestamps monthly, never enter the decay branch, and thus no value of δ can reproduce the sustained decline of Burt's power function (`tests/scan_tie_survival_floor.py`). Accordingly we **do not treat tie-loss rate as a calibration target**, only as a sensitivity dimension. One correction attaches to the Burt citation: the paper's "2.63-year half-life for ties outside the family" is an arithmetic slip (a constant term left unsubtracted); recomputed from the paper's own equation it is 1.63 years[5]. The deep-isolation measure is truncated by the survival floor and therefore enters sensitivity analyses only.

7.4 Modelling choices handled as scans

Three independent literature searches (the third via a multi-source aggregator covering CrossRef/OpenAlex/arXiv) concurred: no authoritative source supports $\delta = 0.06$ or caseload $= 8$—search results that conjured "perfectly matching" numbers (the project encountered one fabricated reference and one fabricated statistic in its first search round) were all rejected and archived. These two parameters are therefore treated as **modelling choices**: δ is scanned over 0.06/0.15/0.25 (scan points chosen from the measured survival curves); caseload in this model effectively controls intervention coverage of high-risk elders (4/8/12 mapping to 40%/80%/100%), a different unit from real-world client rosters—measured average rosters in New York City home-bound elder case management run around 75 clients per worker and are considered overloaded[14]. The Chinese industry standard specifies staffing ratios but no caseload clause[15], and NASW explicitly declines to set a uniform caseload standard[16], so the scan is expressed in coverage terms. Scan conclusions: δ shifts overall levels but does not reorder the interventions; caseload is monotone within the window and converges after withdrawal—coverage affects the in-place effect, not sustainability.

7.5 Root-cause diagnosis of failed runs

Every run failing the validity gates (Section 3.6) was archived and diagnosed, revealing two failure modes: (1) **infrastructure failures**—LLM gateway error storms blacking out whole months of decisions (errors visibly clustered in the trace), handled by rerunning with a higher retry count (an infrastructure parameter, not a protocol change); (2) **interface failures**—LLM calls succeed with zero errors, but after a read-only query returns a false "you cannot contact anyone", the elder stops acting. We earlier classified this as "behavioural silence" and read it as a model trait; a trace review (`ROUTER_INCIDENT_20261021.md`) confirms it is a routing design problem (Section 7.7). The remedy is fixing the router; runs before the fix are judged by the original gate, are not folded into the main results, and the gate is unchanged.

7.6 Model and inference-setting disclosure

The formal batch uses a single model (gpt-5.6-sol) with uniform sampling settings; reasoning-effort level was not recorded by the run script and is disclosed as "not recorded", to be supplemented with a small-sample contrast. Cross-model robustness (gpt-6-astra, seed 1, before vs. after a router fix) is treated as a sensitivity check outside the main results; see Section 7.7.

7.7 Cross-model contrast and routing incident

We swap the focal elders' driving model for gpt-6-astra, keeping profiles, seed, intervention parameters and horizon identical to the seed-1 main-batch runs. Because the contrast straddles a router fix, we make two comparisons that each change one factor: a **router contrast** holds the model fixed (Astra) and compares before vs. after the fix; a **model contrast** holds the router fixed (old router) and compares Sol with Astra. New-router Astra is never compared with old-router Sol. Both are single-seed, single-run descriptive checks, not statistical tests.

Routing incident. Elder models cannot call environment tools directly; they can only ask the environment a natural-language question. When a question is marked read-only, the code-generation router hands only the read-only tools to another model, which writes the answer. The write tools (calling, visiting, asking for help) are registered, but the prompt never asked the generator to distinguish "cannot write in this call" from "this function does not exist", so answers often said "you cannot call or visit anyone". gpt-5.6-sol mostly issues orders without asking first and stays under 10% silence; gpt-6-astra asks more often and treats the answer as a real constraint, then stops acting. This matches the system card's description of the model as more respectful of authorization boundaries[17]. Here the cross-model comparison did what a single model

could not: Sol issued orders directly and masked the router defect, while Astra asked first, stopped, and exposed it. The router was fixed on 2026-10-20: questions about whether a function exists are answered from the registry ("the tool exists; this read-only call cannot invoke it; use readonly=false to act"), and any generated answer still claiming the function is missing is replaced and not cached. The incident timeline and the handling of all eight old-router Astra runs are in Appendix E.

Results after the fix. All four seed-1 conditions ran the full 24 months on the fixed router. Focal-elder silence was 0.0%, 0.0%, 0.0% and 0.7%, with zero LLM call errors, so every run passes the validity gate; old-router none was 11.3% (invalid). Silence essentially disappears, indicating that the high old-router silence came from the environment's answers, not from a model trait; the earlier reading ("less action without visitors") stays withdrawn. The paired casework−none difference has the same direction before and after the fix: suppression during the intervention (old -25.0 pp, new -18.8 pp) and a return close to the no-intervention level after withdrawal (rebound old +25.0 pp, new +15.4 pp). The magnitude shrinks after the fix: old-router none had higher intervention-window isolation (29.6 vs. 26.2) and casework lower (4.6 vs. 7.5). On the fixed router, timebank and platform sit only about 3 pp below none during the intervention; there are no old-router counterparts, so no before/after comparison is made for them.

Model contrast on the old router. On the same old router, Sol's silence was 4.0% (none) and 8.7% (casework), both valid; Astra's was 11.3% (invalid) and 2.0%. The difference is mainly not one of task behaviour but of sensitivity to the same environment defect: Sol issued orders directly and bypassed the wrong answer, while Astra asked first and was stopped by it. Because Astra's none run is invalid, the old-router paired casework−none differences (Sol -17.1 pp, Astra -25.0 pp) are an incident record, not a valid model difference. A clean model contrast would require rerunning Sol on the fixed router, a new paid experiment we did not run (Section 12).

Implication for the validity gate. The silence gate conflates faults, interface issues and model differences; before attributing "doing nothing" to the model, check whether the environment told the agent a wrong capability boundary.

Table 9: Router contrast (A: same model, Astra, before vs. after the fix) and model contrast (B: same old router, Sol vs. Astra); seed 1, single run. Isolation is the window mean (%); intervention = months 7–18, post = months 19–24; rebound = post − intervention (pp). Silence is the share of focal-elder agent-months without a decision (gate 10%).

Router	Condition	Silence	Interv.	Post	Rebound
A. Router contrast (gpt-6-astra)					
Before	none	11.3%†	29.6	30.0	+0.4
	casework	2.0%	4.6	30.0	+25.4
After	none	0.0%	26.2	25.8	-0.4
	casework	0.0%	7.5	22.5	+15.0
	timebank	0.0%	23.3	23.3	+0.0
	platform	0.7%	23.3	22.5	-0.8
Paired casework−none: before			-25.0	+0.0	+25.0
Paired casework−none: after			-18.8	-3.3	+15.4
B. Model contrast (same old router)					
Sol	none	4.0%	25.8	25.0	-0.8
	casework	8.7%	8.8	26.7	+17.9
Astra	none	11.3%†	29.6	30.0	+0.4
	casework	2.0%	4.6	30.0	+25.4
Paired casework−none: Sol			-17.1	+1.7	+18.8
Paired casework−none: Astra			-25.0	+0.0	+25.0

† Exceeds the 10% gate; incident record only. Panel B reflects sensitivity to the same router defect, not task behaviour; the before-fix Astra rows are the same data in A and B. Completion evidence for the fixed-router runs is the run-chain log plus the program's success record (this cohort did not use DONE flags).

8 LLM Call Accounting

Resource reporting for multi-agent LLM experiments must include runs eliminated by quality gates. We report only trace-measured call counts and no dollar estimates (traces do not record per-call token usage, so any amount would rest on assumptions). The main batch and the old-router contrast arms made 45,901 LLM calls: 22,987 on valid runs, 17,778 on runs archived as invalid (44% of the main batch), and 5,136 on the old-router cross-model arms. The four fixed-router gpt-6-astra conditions add 5,307 calls with 0 call errors (Section 7.7); the combination arms D and E add 2,849 calls with 9 call errors, and the combination schedule loop's designer made 6 calls (Section 6.3). Measured density in the focal layer is about 13.8 calls per focal elder-month. Here the layered design pays off concretely: the rules-layer scans (20 seeds × 45 configurations) and the harness-loop evaluations make no LLM calls, and the loop designer makes one call per round; running the same scans through the LLM layer would exceed the main batch's call volume by an order of magnitude. The ledger is `tests/llm_calls_ledger.json`.

9 Core Conclusions and Innovations

Core conclusions. (1) **What remains when a service ends**: social-worker visits activate family only while they run; the peer contact left after mutual-aid pairing is supplied by the pair-persistence parameter, and LLM elders do not maintain pairs; volunteer matching leaves no everyday relationship; all three services leave only a help-seeking channel to former providers (Section 5.3). Which quantity governs each layer's persistence, and what should be measured in practice, is the main output of this paper. (2) Generating profiles from survey distributions is necessary but not sufficient: with anchored profiles alone, outputs are no closer to reality than with ordinary random profiles, and the error comes mainly from loneliness dynamics;

once dynamics are calibrated under preregistration and accepted on held-out targets, survey-anchored profiles beat random ones on output (Section 7.2). (3) Under this model's parameters, the in-window ranking is social-worker visits > mutual-aid pairing > volunteer matching; the direction is robust to the profile source while the magnitude depends on inputs; social-worker visits fall back to the never-treated level after withdrawal, and the post-withdrawal advantage of mutual-aid pairing is set entirely by the pairing-persistence parameter (Section 5). (4) Under rules-layer mechanics, entry timing shows up as a delay cost accumulating linearly by month rather than as a "missed critical window" (Section 6). (5) Running social-worker visits and mutual-aid pairing together is sub-additive; "social workers open the way, mutual-aid pairs carry on" holds only if referral by a social worker actually makes pairs last longer, a hand-over strength the model cannot supply and that must be measured in the field (Section 6.3).

A methodological finding. The cross-model comparison exposed an environment defect that a single model could not: Sol issued orders directly and masked the router's habit of phrasing "cannot write in this call" as "cannot contact anyone", while the more boundary-respecting Astra asked first, stopped, and exposed it (Section 7.7). "Doing nothing" in LLM social simulation should be attributed to the model or the simulated person only after checking what the environment told the agent.

Innovations. (1) Turning "is the simulation merely an echo of its rules?" into a testable question: a result-source ledger item by item, and zeroing and scanning experiments that take key conclusions apart; (2) pushing survey anchoring from "do the profiles look right?" to "are the outputs right?": a preregistered ablation plus train/held-out calibration, reporting the negative pre-calibration result as well; (3) a layered architecture that makes large scans free of API cost while keeping the behaviour and reasons of an LLM focal layer; (4) organising the comparison around the social-work, social-security and social-governance channels so that simulation questions map onto practice decisions.

10 Discussion: Implications for Practice

(1) Inner layer: social-worker visits need a built-in exit plan. The social worker is an injected active node, so the strongest in-window effect and the post-withdrawal fall-back are two faces of one mechanism; the family contact it brings exists only while the service runs (Section 5.3). Exit design—gradual tapering and referral into family and reciprocal networks—should be a standard component of outreach. What must be measured is whether social work actually changes the strength of family ties, and whether it crowds them out.

(2) Middle layer: the sustainability of mutual-aid pairing rests on an unmeasured quantity. In the model its post-withdrawal advantage comes entirely from the pair-persistence parameter and vanishes when it is zeroed, and LLM elders hardly ever contact pair partners themselves. "Social workers open the way, mutual-aid pairs carry on" is therefore only a hypothesis; whether it holds depends on how often real pairings persist after a programme ends, which the time-bank literature has not measured[3]. The combination experiment (Section 6.3) adds that rolling out both services at once covers the same elders twice, and a carry-on advantage can appear only if the hand-over itself is designed and measured.

(3) Outer layer: volunteer matching is worth its emergency reach, not lasting ties. Volunteer ties remain after withdrawal but are used only when elders face an event; whether weak ties turn into familiar ones is not modelled and its conversion rate must be measured. Volunteer matching has a weak total effect and is insensitive to timing, so optimising its timing adds little.

(4) What remains after withdrawal is a help-seeking channel. After all three services elders contact former providers only to ask for help; removing that channel raises post-withdrawal isolation. Keeping a reachable contact when a service ends may be cheaper than extending the service, but whether real providers still respond and whether elders are willing to ask must be measured.

(5) Waiting time is a design variable. Under rules-layer mechanics, delay cost accumulates monthly (about +0.15pp per month for social-worker visits). This magnitude rests on the assumption that an intervention produces contact as soon as it starts; real start-up lags and coverage ramps would change it.

All of the above are model projections for generating testable practice hypotheses; **we claim no causal effects in real populations**.

11 Conclusion

We propose and demonstrate a layered agent-based simulation method for research on social-work interventions: an LLM focal layer embedded in a rules-layer network, survey-anchored profiles validated on held-out targets, and externalised mechanism parameters, together with preregistration, validity gates, root-cause classification, LLM-call accounting and a result-source ledger. The case of older adults living alone asks where relationships go when a service ends: family activation does not last, peer pairing depends on a persistence rate not yet measured, weak ties do not settle, and all three services leave only a help-seeking channel; each projection can be tested under explicit parameters by zeroing, removal or scanning and turned into a quantity to measure in practice. Every number updates automatically from the scripts; we invite reproduction per the appendix.

12 Limitations and Ethics

Limitations. (1) $n = 3$ seeds ($n = 2$ for the platform arm) support only descriptive contrasts; (2) profiles match only survey marginals; loneliness dynamics were calibrated post hoc in the rules layer only (selected point on the grid boundary), contact frequencies and tie decay remain modelling choices, the tie-survival curve fails the Burt check, and the LLM main batch ran on pre-calibration dynamics; (3) intervention intensities and inputs are modelling choices not aligned with literature effect sizes or service volumes, so the ranking among interventions is set by parameters; (4) the primary outcome is a monthly contact snapshot and loneliness follows a steady-state formula, so neither is an independent outcome; (5) macro results come mainly from six LLM elders and are sensitive to model behaviour and silence, and the withdrawal window is only six months; (6) the rules-layer timing scan and the harness loop search only under fixed rules and do not modify model parameters; (7) the main batch ran on the pre-fix router: focal elders who queried the environment could receive a false "cannot contact" answer, so although valid Sol runs stay at 4.0–10.0% silence, some silent or low-action months may stem from it, and focal-layer results should be read accordingly; Sol was not rerun on the fixed router, so there is no clean post-fix model contrast; (8) the LLM layer of the combination experiment has only the seed-1 D and E arms and is descriptive, so inference on interaction and hand-over rests on the rules layer and the explicit hand-over parameter; (9) the model does not represent crowding-out of family by social workers, reciprocity running on its own without credits, or weak ties becoming familiar ones; old service ties stay on the ledger at withdrawal, which produces the post-withdrawal help contacts; LLM-layer help requests do not record whether a deep talk occurred. Follow-up work is listed in the parameter appendix and `DATA_MACHINE_RUNBOOK.md` in the repository.

Ethics. All agents are simulated entities; no real individuals are involved. Survey data enter only as aggregate distributions from public microdata. The LLM-generated "elders' rationales" are mechanistic probes, not real elders' voices, and the paper avoids quoting them anthropomorphically. Because simulation conclusions could be misused as direct grounds for resource allocation, we repeatedly stress their hypothesis-generating character.

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A Reviewers' Concerns and Corresponding Sections

Table 10: Review criteria and questions raised before the finals, with where this paper answers them.

Concern	Answer	Where
Social-science value	Addresses post-withdrawal and entry-timing questions; organised around three policy channels	Sec. 1, 10
Novelty of the question	Makes "is the simulation an echo of its rules?" testable	Sec. 9
Rigour of method	Preregistration, gates, root causes, call accounting, result-source ledger	Sec. 3, 7, 8
Findings and theory	Anchoring needs dynamics calibration; cross-model comparison exposes environment defects	Sec. 9
Impact and extensibility	Parameterised sandbox; transferable checklist	App. C
Why better than random profiles	Preregistered ablation: closer inputs; after calibration 4.3 vs. 7.3 pp output error	Sec. 7.2
Why these three interventions	Convoy-model layers and the social-work / social-security / social-governance channels	Sec. 3.4
Social-work or sociology framing	Social-work practice as the main line; theory from social support and the convoy model	Sec. 1, 2
Timing value and harness iteration	Delay cost is a rules-layer directional estimate; the loop is used for dynamics calibration, no self-evolution claimed	Sec. 6, 7.2
Cross-model comparison	Old-router differences stem from a routing defect, fixed and documented	Sec. 7.7
Defined vs. experimental indicators	Result-source ledger, item by item	Table 7

B Reproducibility Checklist

- **Code and data**: AgentSociety2 repository `examples/v2/elder_support/`; environment module `custom/envs/elder_support_env.py`.
- **Formal batch**: `run_batch.sh 24 1 2 3` (automatic checkpoint resumption; validity gates and archiving built in).
- **Analysis and figures**:

`tools/compare_conditions.py` → `comparison.json`; `tools/make_figures.py` (each figure ships with a same-name CSV); `tools/gen_paper_numbers.py` (the sole source of every number in this paper).

- **Mechanism scans and loop**: `tests/scan_timing.py`, `tests/scan_post_persist.py`, `tests/scan_tie_survival_floor.py`, `tests/scan_combo.py`, `tools/harness_loop.py`, `tools/harness_loop_combo.py` (preregistered headers + per-round JSONL audit logs).
- **Call counts and diagnosis**: `tests/llm_calls_ledger.json`, `tests/silence_diagnosis.json`.
- **Calibration documents**: `data_pipeline/out/calibration_targets.md`, `references_verified.md` (including rejected entries and verification methods).
- **Live dashboard**: https://agentsociety.shangdian.me (batch progress, conclusion cards, mechanism evidence, cost and failure diagnosis).

C A Transferable Checklist: Scientific Discipline for LLM Social Simulation

The table below compresses the discipline practised in this paper into checklist items that other LLM social-simulation studies can apply directly, each with the place where this paper implements it.

#	Item	Where in this paper
1	Hypotheses, primary/secondary outcomes, seeds and validity gates written into versioned scripts before the batch	`run_batch.sh` header; Section 3.6
2	Gates distinguish infrastructure-type from interface-type failure; check environment replies before blaming the model	Section 7.5; Section 7.7
3	Failed runs archived, not deleted, each with a root cause; no "run until it passes"	`*.invalid-*` directories; `silence_diagnosis.json`
4	Input distributions anchored dimension by dimension to real surveys, with definitional gaps and uncovered parts stated	Section 3.5
5	At least one output-side external-validity check, failures reported	Sections 7.2 and 7.3
6	Every headline result labelled with its source (rules, parameters, or LLM-layer description)	Table 7
7	Hybrid architectures decompose each layer's contribution to the macro outcome, with silence-excluded figures	Table 5
8	Nature of indicators stated: snapshot vs. dynamic; which one effect sizes are read from	Section 3.6
9	Baseline silence rate without intervention reported per model as a selection statistic	Section 7.7
10	Call counts include invalid and comparison runs; measured counts only	Section 8
11	All numbers in the text generated by script, none transcribed by hand	`numbers.tex`
12	Propose–evaluate loops have a preregistered search space, round and resource caps, and a per-round audit log	Section 6.2

D Validity-Gate Adjudication Rules

A run is valid if and only if: the DONE flag exists; monthly metrics for months 0–24 are complete; decision records are non-empty; the six focal elders' combined silence rate is $\leq$10% (strictly greater than 0.10 fails); and the LLM call error rate is $\leq$5%. In-

valid runs are archived under timestamped directories (`*.invalid-<timestamp>`) with all traces and decision logs preserved for diagnosis—never deleted.

E Routing incident timeline

- 2026-09: gpt-6-astra none seed 1 twice, silence 10.7% and 11.3%, both invalid; casework seed 1 at 2.0%, valid. At the time this was misread as a model trait.
- 2026-10-19: five more Astra runs on the old router, silence about 31–87%, never a complete four-condition set; the same "cannot contact" answer also appears in the gpt-5.6-sol platform seed 2 re-run (60.7% silence).
- 2026-10-20: the read-only router was diagnosed and fixed (Section 7.7); a 2-elder × 2-month smoke test showed real calls and visits and is not counted as a valid run.
- 2026-10-21 to 10-22: the four conditions ran in sequence on the fixed router with the seed-1 main-batch configuration. Casework exited abnormally about one minute after its first start and completed after one restart; the other conditions completed on the first attempt.

All old-router Astra runs are archived unchanged; Table 9 cites only the September 2026 seed-1 none and casework runs.

F Parameter Table and Rationale

Table 11 lists the parameters of the calibrated profile generator and the environment module. Environment and intervention parameters are centralised in `DEFAULT_DYNAMICS` and `INTERVENTION_DEFAULTS` and can be overridden by configuration; values shown are the defaults used by the main batch. "Survey" means sampled directly from a survey distribution; "mapping" maps survey categories to model values or thresholds; "assumption" means an author-set modelling choice without a point estimate from data or literature. "Scan" lists only completed sensitivity scans.

Table 11: Parameter table (s: tie strength; e: extroversion; h: health).

Parameter	Value	Basis	Scan
Profile generation			
Age, sex, self-rated health, loneliness, living children, frequent visits	Unweighted distributions of CLHLS 2021 living-alone subsample ($n = 1707$)	Survey	No
Child distance	Local / same city / farther = 28/36/36	Survey (CHARLS 2015)	No
Numerical health / loneliness categories	Health centres 0.88–0.20; loneliness centres 0.85–0.15; within-category SD 0.06	Mapping	No
Child contacts per month	Frequent visitors 5/3/1.8, others 1.5/0.8/0.4 (local / same city / farther), times $U(0.7, 1.3)$	Assumption	No
Extroversion	$N(0.5, 0.2)$ truncated to $[0.05, 0.95]$	Assumption	No
Neighbour / friend ties	Count 0–4 with weights 10/25/35/20/10, +1 if $e > 0.7$; strength $N(0.5, 0.18)$	Assumption	No
Environment dynamics			
Tie decay δ	0.06; delete if $s < 0.05$; child-tie initial strength 0.8	Assumption	δ 0.06–0.25
Monthly events	Illness 0.10, fall 0.03, neighbour moves 0.02; annual-age factor +3% after 70; illness times $(2 - h) \times 0.6$; health loss 0.05/0.08	Assumption	No
A contact becomes emotional	In-person $0.25 + 0.35s$; phone $0.10 + 0.35s$	Assumption	No
Help seeking	Success $0.15 + 0.8s \times$reachability (children: 1.0/0.7/0.15); success s+0.08, loneliness −0.06; failure s−0.05, loneliness +0.05	Assumption	No
Rules-layer monthly behaviour	Expected child contacts = monthly frequency $\times(0.5 + 0.5s)$; neighbour interaction $e(0.2 + 0.45s)$; no-help probability $0.25 + 0.5(1 - e)$; at most 2 targets	Assumption	No
Loneliness steady state	$0.90 - 0.12\min(\text{emotional}, 4) - 0.05\min(\text{ties}, 5) - 0.25h + 0.10(1 - e)$; halfway to target monthly	Assumption	relaxation 0.2–0.5; base 0.80–0.90
Monthly loneliness shock	Main-batch default 0; post-hoc calibration selected $N(0, 0.20)$	Assumption (train/hold-out calibration)	0–0.20
Lonely threshold	$\ell \geq 0.6$, mapped to CLHLS b38 often/always	Mapping	No
Deep isolation (diagnostic)	3 consecutive months without emotional contact and no instrumental support this month	Assumption	No
Focal-elder tools	Call s+0.04; visit s+0.05; activity loneliness −0.04, new-friend probability $0.35 + 0.4e$, strength 0.25	Assumption	No
Risk score	Months without emotional contact 2.0; $(1 - h)$ 1.5; tie deficit below 3: 1.0; events 0.8; loneliness 1.0	Assumption	No
Interventions (direction from literature; numerical values are modelling choices)			
Intervention window	Months 7–18	Design	Start month
Social-worker visits	2 workers × caseload 8; tie strength 0.4; visit emotional 0.8, s+0.05; family reconnection 0.5, emotional 0.4, s+0.05	Assumption	caseload 4/8/12
Mutual-aid pairing	Helper $h \geq 0.6$, age ≤ 78, max 2 recipients; monthly visit 0.85; strength 0.35; recipient emotional 0.6, instrumental 0.5, s+0.06; helper emotional 0.4, s+0.04	Assumption	No
Pairing after withdrawal	post_persist 0.5; when persistent: recipient emotional 0.5, s+0.03; helper emotional 0.3, s+0.02	Assumption	post_persist 0–1
Volunteer matching	Event this month or 2 months without emotional contact; coverage 0.6; strength 0.2 capped 0.35; emotional 0.3; s+0.03	Assumption	No

Initial loneliness is sampled from the survey, but the monthly steady-state update halves its influence each month; only 1/8 remains after three months. Simulated loneliness therefore comes mainly from the assumed steady-state equation. Initial rules-layer scans used placeholder profiles; calibrated-profile reruns on 2026-10-21 kept the same directions. The formal LLM batch uses the default dynamics above; post-hoc calibrated values are not written back into its results.